After Mid

☐ Memory Interfacing

☐ Chip Identification:

→ 2 memory element : EPROM , RAM

1. EPROM : IC 27XXX , IC 27XX

2. RAM : IC 61/62 XXX

EX-1:

IC $\boxed{27}\boxed{512}$

1st two digit memory Type — Size of memory in Kbits.

→ Memory size = 512 Kbits = $\frac{512}{8}$ KB = 64 KB

→ That's memory is EPROM 64 KB memory.

EX-2

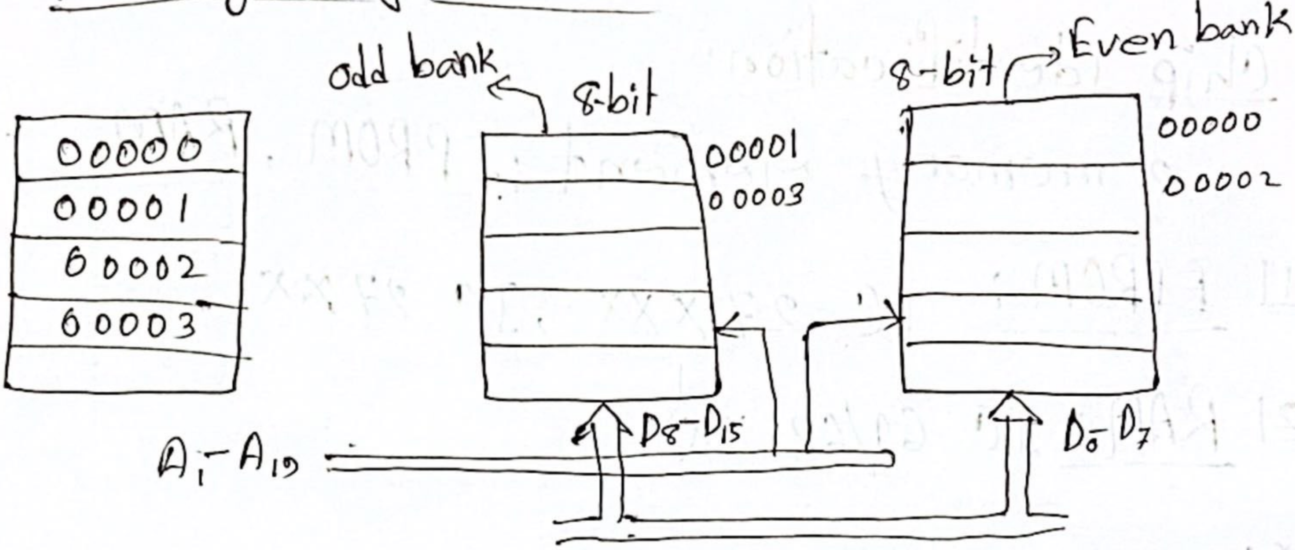
IC $\boxed{62}\boxed{64}$

1st two digit is memory type — Size of memory in kbits.

→ Memory size = 64 Kbits = $\frac{64}{8}$ KB = 8 KB

→ That's memory is RAM 8KB memory.

Memory Organization:



=> Absolate/ Full decoding chip selector এর সবগুলো bit নিয়ে কাজ করতে হবে।

=> Partial ~~de~~/ leaner decoding chip selector এর 2/3 টা bit নিয়ে কাজ করতে হবে।

=> Memory ২টা থাকলে অথবা বিজোড় সংখ্যক memory থাকলে 2 জোড়া করে কাজ করতে হবে অর্থাৎ pair করে করতে হবে।

=> If Memory Size Same না এবং Memory Same হলেও। অথবা Memory size Same কিন্তু memory same না হলে pair করতে পারবো না। ~~[illegible]~~ same memory and same size হলেই 2টি memory pair করে কাজ করতে পারবে।

P-1: Interface two 4K X8 EPROM and 4K x8 RAM chip with 8086.

Sol^n: Given that,

Memory size of

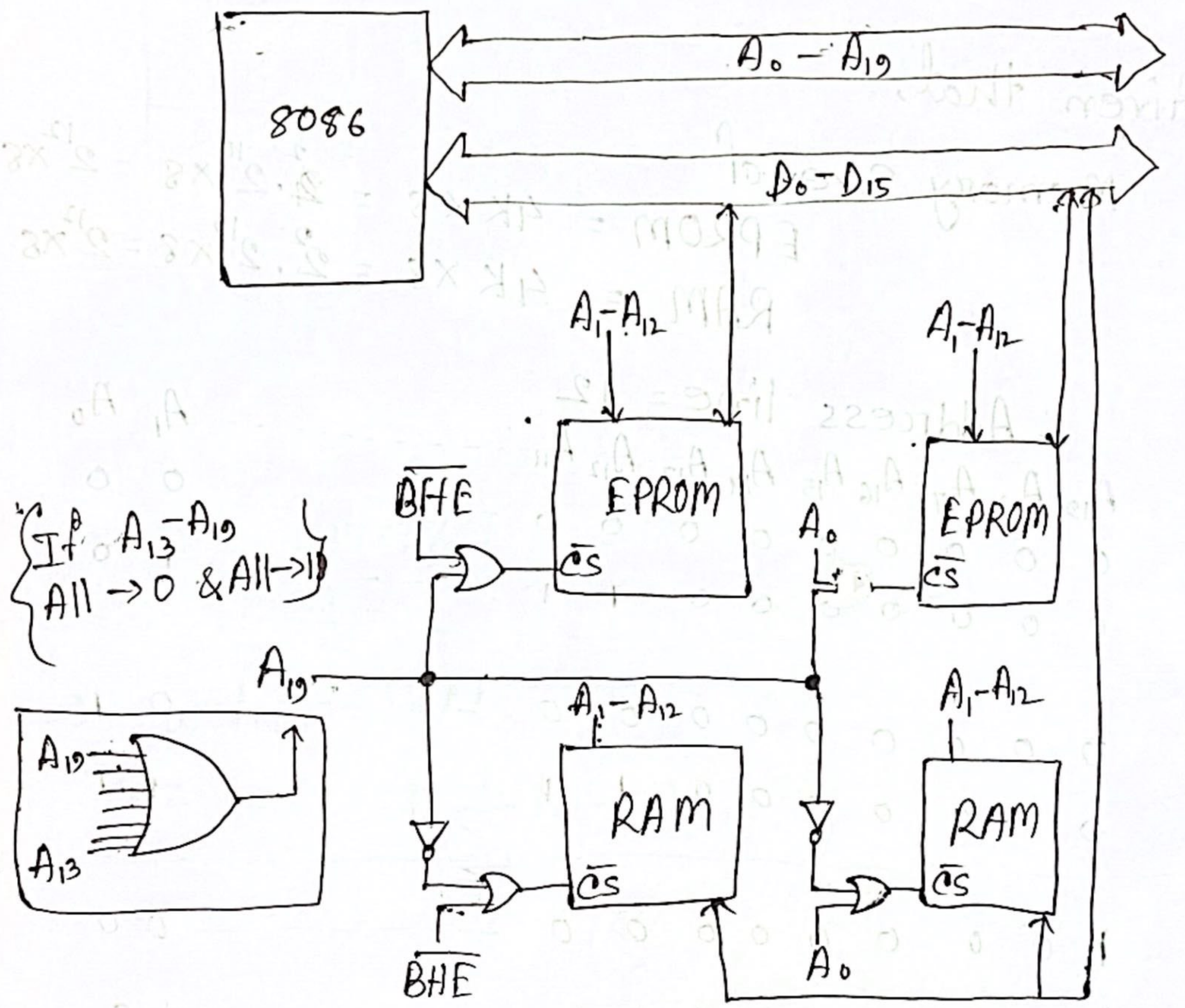
$$EPROM = 4K \times 8 = 2^2 \cdot 2^{10} \times 8 = 2^{12} \times 8$$

$$RAM = 4K \times 8 = 2^2 \cdot 2^{10} \times 8 = 2^{12} \times 8$$

$\therefore$ Address line = 12

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	------	A_1	A_0
EPROM-I	0	0	0	0	0	0	0	0	0	------	0	0
	0	0	0	0	0	0	0	1	1	------	1	0
EPROM-II	0	0	0	0	0	0	0	0	0	------	0	1
	0	0	0	0	0	0	0	1	1		1	1
RAM-I	1	0	0	0	0	0	0	0	0	------	0	0
	1	0	0	0	0	0	0	1	1	------	1	0
RAM-II	1	0	0	0	0	0	0	0	0	------	0	1
	1	0	0	0	0	0	0	1	1	------	1	1

Diagram;



8086
$A_0 - A_{19}$
$D_0 - D_{15}$
$A_1 - A_{12}$
EPROM
$\overline{BHE}$
$\overline{CS}$
A_0
If $A_{13} - A_{19}$
All → 0 & All → 1
A_{19}
RAM
A_{13}

P-2: Interface 32KB of RAM memory to the 8086 microprocessor system using absolate decoding with the suitable address.

Soln:

Given that,

Total RAM = 32 KB

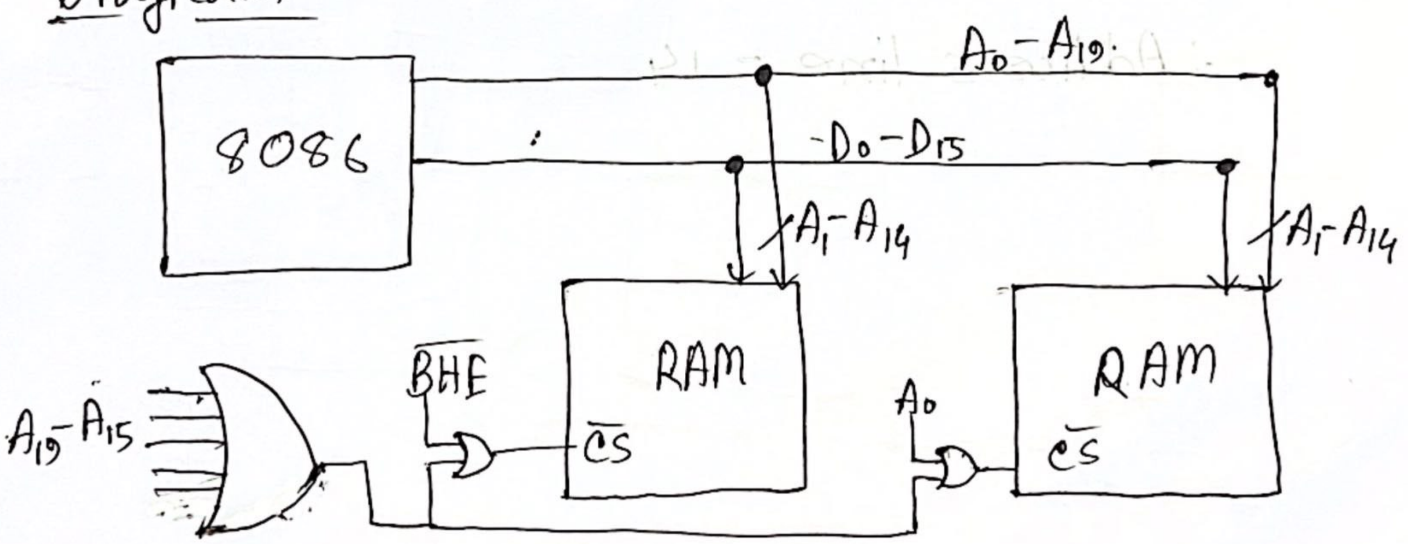
$\therefore$ Even and odd Bank $= \frac{32}{2}$ KB $= 16$ KB

$= 2^4 . 2^{10} \times 8 = 2^{14} \times 8$

$\therefore$ Address Line = 14

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}		A_1	A_0
								----	0	0
RAM-I	0	0	0	0	0	0	0		1	0
	0	0	0	0	0	1	1			
RAM-II	0	0	0	0	0	0	0	----	0	1
	0	0	0	0	0	1	1		1	1

Diagram



P-3: Interface 32K word of memory to the 8086 microprocessor system. Available memory chips are 16K×8 RAM use suitable decoder for generating chip select logic.

Soln:

we know that,

word = 16 bits = 2B

Required memory

32K words = 32K × 2 B

= 64KB

∵ Eeach RAM chip = 16K×8 = 16KB

∴ so, nomber of chips required $= \frac{64KB}{16KB}$

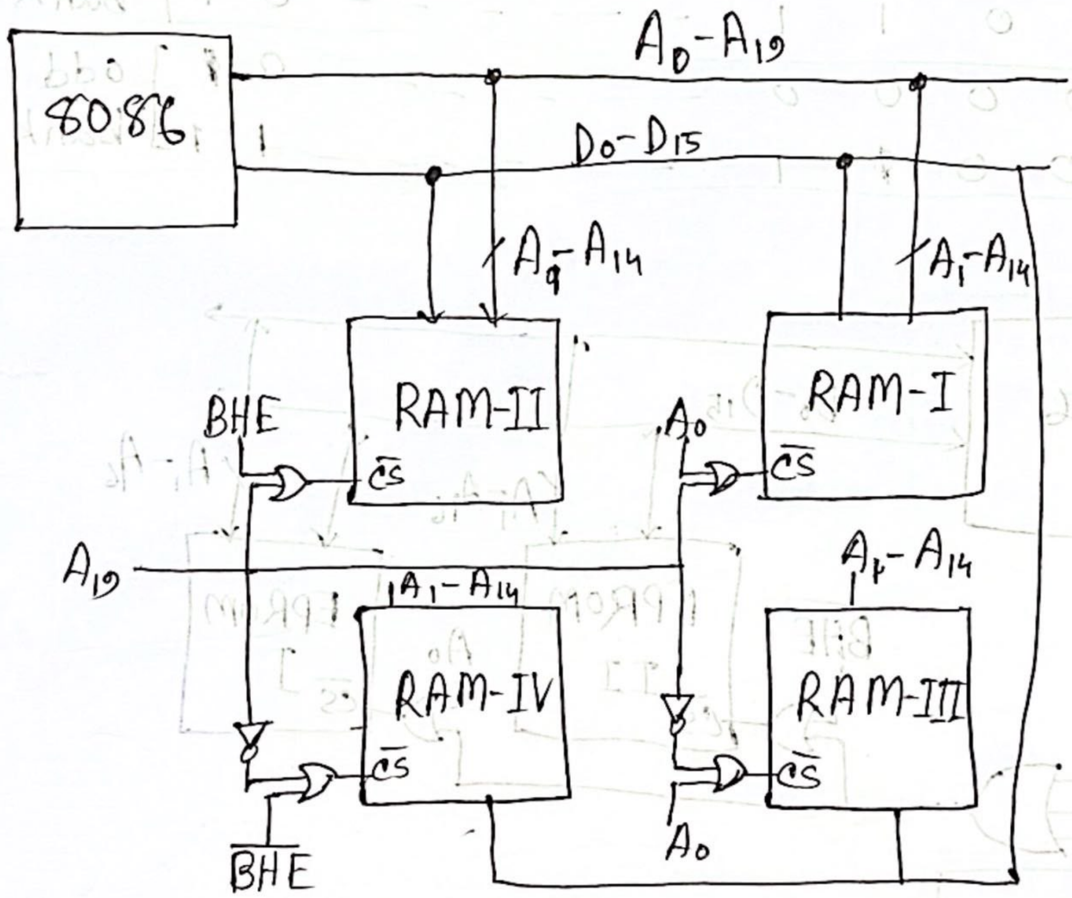
= 4 chips

RAM → $16KB = 2^4 . 2^{10} \times 8 = 2^{14} \times 8$

∴ Address line = 14

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	---	A_1	A_0
RAM-I	0	0	0	0	0	0	0	---	0	0
	0	0	0	0	0	1	1	---	1	0
RAM-II	0	0	0	0	0	0	0	---	0	1
	0	0	0	0	0	1	1	---	1	1
RAM-III	1	0	0	0	0	0	0	---	0	0
	1	0	0	0	0	1	1	---	1	1
RAM-IV	1	0	0	0	0	0	0	---	0	1
	1	0	0	0	0	1	1	---	1	1

Diagram:



P-4: Interface microprocessor 8086 with two number of IC-27512 chip.

Solⁿ:

Given that,

2 number of IC-27512 → EPROM

$\therefore$ memory size $= \frac{512}{8}$ KB $= 64$ KB

$= 2^6 \times 8$

$\therefore$ Address line $= 16$

$= 2^{16} \times 8$

	A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	----	A_1	A_0	
EPROM -I	0	0	0	0	0	----	0	0	Even bank
	0	0	0	1	1	----	0	1	
EPROM -II	0	0	0	0	0	----	0	1	odd bank
	0	0	0	1	1	----	1	1	

Diagram